

Knobula

Monumatic

Instruction Manual

Version 1.1.4



Contents

Introduction	3
Getting Started	4
Power	4
Midi	4
Midi Channel Select	4
Midi Polyphonic Expression (MPE) Operation	4
CV Gate	4
Stereo Output	4
Envelope	5
Attack (volume)	5
Decay Sustain (glide)	5
Release -chord select	5
Push Buttons	6
Trigger	6
Shift	6
Mono/Poly (chorus)	6
Poly Midi	6
Poly CV & Gate	6
Poly Gate Only	6
Monophonic Midi	6
Monophonic CV & Gate	6
Monophonic Gate Only	7
Arpeggio Mode	7
Chord Mode	8
Change Active Chord - with a Knob	8
Change Active Chord -Via Midi	8
Store Custom Chord	8
Audition Chord	8
Filter	9
Frequency (voice pan)	9
Resonance (midi channel)	9
Drive	9
Depth	9
Key Follow	10
Type	10
LP24	10
AP	10
Vowel	10
Low	10
Notch	10

High	10
Oscillator	11
Detune (temperament)	11
Pitch (fine tune)	11
Oscillator Mode Selector	11
Oscillator Mode Descriptions	12
Effects	13
Reverb (reverb time)	13
Chorus	13
Inputs and Outputs	14
Gate In	14
Filter	14
CV A	14
CV B	14
Midi In	14
Assignable CV	15
Assigning a CV to a Control Knob	15
Assigning Oscillator Mode	15
Assigning Filter Frequency	15
Assigning Chord Select	15
Assigning Pitch Knob	15
Temperament	16
Updating the Firmware	16
Display Current Firmware Version	17
Update Procedure	17
Firmware Trouble Shooting	18
Monumatic No longer working after firmware upgrade	18
Monumatic Not showing in Connect Menu	18
CV Calibration	19
Automatic Calibration	19
Midi CC Calibration	19
Factory Reset	19
Appendix A	20
Midi CC Chart	20
WARRANTY	21

Introduction

Monumatic is designed from the ground up to be the ultimate compact poly synth module for eurorack. Taking its inspiration from the original Poly Cinematic, the Monumatic refines the concept of the hands-on compact polysynth by extending the CV functionality, adding more oscillator modes and more filter types as well as a host of other features that expand the sonic possibilities of a 12hp synthesizer voice.

Starting with a whole new hardware re-design and borrowing from recent innovations in its sibling modules, Monumatic offers 16 oscillator modes, dedicated bipolar V/oct and Filter CV inputs plus two additional CV inputs that are instantly assignable to any knob on the front panel including the oscillator mode selector. Six switchable filter types include 24db/oct Butterworth filter and a 4 stage All Pass phaser, also a filter overdrive switch allows for harmonically rich overtones and gnarly lead sounds. Switchable monophonic and polyphonic operation at the touch of a button, and MPE/Poly aftertouch is also fully implemented. Whether as a lead oscillator in a giant patch or as a stand alone polysynth, Monumatic is the most essential, versatile module in any modern eurorack case.

Getting Started

Power

Connect the unit to a standard Eurorack +/- 12v power supply using the ribbon cable provided.

Midi

Connect the trs-A cable provided into the Midi input and the other end to a suitable controller or sequencer.

Midi Channel Select

To select a midi receive channel hold Shift and turn the filter Resonance control. Values available are All (Default), then 1 through 16. The Trigger button LED will flash each time the value changes.

Midi Polyphonic Expression (MPE) Operation

To use MPE controllers that send note data per channel, set the midi channel to Omni (hold Shift and turn resonance down to zero) and press the Mono button on (Mono mode - led glows red). Monumatic assigns one voice per midi channel on channels 1 through 8. Polyphonic pitch bend, filter Brightness (CC74) and aftertouch can be sent to individual voices. Aftertouch is assigned to the sustain level of the envelope.

CV Gate

For CV Gate operation connect a Gate or Trigger signal to the Gate Input and a 1v/Oct source to the V/Oct Input.

Stereo Output

Connect the left and right outputs to eurorack audio inputs.

Envelope



Attack *(volume)*

Adjusts the fade in time for each note.

Hold Shift to adjust the output level of the whole device.

Decay | Sustain *(glide)*

Changes the decay time, the time taken for the sound to die away while the key is pressed. Turning the knob past the midpoint sets the sustain level and the Release knob defines the decay time to reach that level.

Hold Shift to adjust glide time. Unlike portamento, glide time is constant no matter how far the pitch has to change - it will always take x milliseconds to reach the intended pitch. This behaviour is similar to the slide feature on the Roland TB-303.

Release -chord select

Adjusts the time taken for the sound to die away after the key is released. Turning the control fully clockwise will hold the note indefinitely.

Hold **Trigger** and turn the knob to change the active chord memory.

Push Buttons

Trigger

Press to manually play a chord from the chord memory.

Shift

Holding Shift while turning a knob enables additional parameters to be controlled. These are (*Labelled in brackets*) on the front panel. After adjusting a Shift value the knob is disabled temporarily until it is moved again.

Mono/Poly (*chorus*)

When the LED is dark, Monumatic is in polyphonic mode.

Hold Shift to enable/disable the stereo chorus. Chorus speed depends on Detune position.

Poly Midi

Up to 8 voices are assigned to notes coming from the Midi input.

Poly CV & Gate

If the module is being played by CV and Gate each successive note will overlap the last according to the release setting, enabling layered polyphonic arpeggio effects.

Poly Gate Only

If only the Gate input is connected, Monumatic will play the current chord in the chord memory. This chord can be constantly updated over midi while being triggered by the Gate input.

When the red mono/poly LED is lit, the module is in *monophonic* mode.

Monophonic Midi

The red LED glows in mono mode, each note played will be truncated by newer notes and the envelope will be retriggered. Pressing the Trigger button will engage Chord Playback where a stored chord is assigned to the pitch and velocity of the note played.

Monophonic CV & Gate

Notes will play in legato mode much like a traditional analog mono synth, pressing Trigger will engage Chord Playback mode where a stored chord is assigned to the pitch of the note played.

Monophonic Gate Only

If only the Gate input is connected, and it is in Mono mode (red led on) Monumatic will play each note of the current chord in a sequence. This chord can be constantly updated over midi while being arpeggiated by the Gate input.

Arpeggio Mode

See above.

Chord Mode

Whilst in monophonic mode press the Trigger Button to engage chord playback mode, the trigger led will glow white. In Chord mode a chord of up to 8 notes is assigned to the pitch and velocity of the note played.

Change Active Chord - with a Knob

To change the chord hold down Trigger and turn the Release knob to audition each chord memory.

Change Active Chord -Via Midi

Use the modulation control, when the memory slot changes the Trigger led will flash.

Store Custom Chord

To store new chords enter polyphonic mode and using a midi controller/keyboard play a chord - Monumatic will memorise the last chord you played and assign it to the currently active chord memory slot.

Audition Chord

Press the Trigger button to audition chords in either monophonic or polyphonic mode.

Filter



Frequency *(voice pan)*

Adjust the cutoff frequency of the selected filter type. In Vowel mode this changes the vowel sounds through Ahh, Ay, Eeee, Eye, Ohh, U.

Hold Shift to increase Voice Panning, each note is panned to a unique position in the stereo field.

Resonance *(midi channel)*

Adjusts the resonance emphasis of the current filter type. In Vowel mode this control changes the formant.

Hold shift to change the current midi channel, values range from All Channels / MPE mode, then 1 through to 16, the Trigger button will flash each time the value changes.

Drive

Signals are overdriven as they exit the filter for more harmonically rich overtones emphasised at the cut off frequency.

Depth

Adjusts the amount that the envelope modulates the cutoff frequency of the filter.

Hold Shift to modulate the oscillator pitch with the envelope.

Key Follow

The note frequency influences the cut-off frequency of the filter. When set to full, the filter frequency is proportional to the note frequency.

Type

Use the 3-way toggle switch and the slide switch to select between the six different filter types:

LP24

24 dB/Oct Butterworth low pass filter.

AP

4 stage All Pass filter for phase shift effects.

Vowel

Vowel filter for creating vocal effects. Turn filter frequency to morph between different vowel sounds - Ahh, Ay, Eeee, Eye, Ohh, U.

Turning the Resonance will alter the formant from monster to chipmunk at the extremes with male and female humans in the middle.

Key Follow naturally correlates the formant with the change in pitch.

Low

12 dB/Oct Low pass filter, frequencies below the cut off are removed.

Notch

12 dB/Oct Notch filter, frequencies near the cut off are removed.

High

12 dB/Oct high pass filter, frequencies above the cut off are removed.

Oscillator



Monomatic contains up to 56 oscillators across its 8 voices. Each oscillator mode represents a combination of several parameters and modulation sources that form the basis of a subtractive synthesis patch.

Detune (*temperament*)

Depending on the oscillator mode, the Detune control will either detune each oscillator from Unison to Fifth to Octave, or it will control different characteristics within the waveform itself. *Hold Shift to change the temperament from equal to just (midpoint) to unjust (fully clockwise)*
See below for explanation of temperaments.

Pitch (*fine tune*)

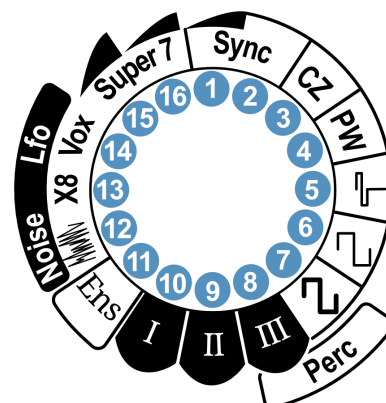
Adjust the pitch up or down an octave in coarse increments of one semitone.

Hold Shift for fine tuning.

Note: When using the Vowel filter, the filter's central formant frequency will follow the pitch setting.

Oscillator Mode Selector

16 different oscillator modes are available to choose from. Each mode represents both a different waveform combo and a different internal patch or oscillator model. Depending on the mode there are different ways the detune control will affect the sound, generally the oscillators will be detuned between sweet spots of unison/fifth/sub-octave, and applicable LFO speed will increase between these sweet spots giving a more dissonant timbre. Certain



if

modes are only single oscillator so the detune will only change the waveshape.
Starting from 12 O'clock on the Mode selector and moving clockwise the modes are as follows:

Oscillator Mode Descriptions

Name	Description	Detune Function
1. Sawtooth Sync	Osc 2 is synced to start of Osc 1. Sub oscillator modulates Osc 2 octave.	Osc 2 freq ramped up to 4 octaves for each zone of Osc 1. Osc 1 Zones - Unison 5th and Suboctave.
2. Square Sync	Osc 2 is synced to start of Osc 1	Osc 2 freq ramped up to 4 octaves for each zone of Osc 1. Osc 1 Zones - Unison 5th and Suboctave.
3. CZ	Casio inspired phase distortion oscillator	Sawtooth - Square - Sawtooth + Sub Phase distortion changes between zones.
4. PWM	Single Osc with adjustable pulse width	Unison - 5th - Suboctave. Pulse width changes between detune zones.
5. Pulse	3 Osc Pulse	Unison - 5th - Suboctave. Pulse width changes between detune zones.
6. Square	3 Osc square	Unison - 5th - Suboctave. Pulse width changes between detune zones.
7. Square Perc	3 Osc square. Osc 3 has ring-modulated short decay envelope	Unison - 5th - Suboctave. Pulse width changes between detune zones.
8. Organ III Perc	Triple Drawbar Jazz Organ with "percussion" 5th	Unison - 5th - Suboctave. Leslie effect increases between detune zones.
9. Organ II	Triple Drawbar Jazz Organ	Unison - 5th - Suboctave. Leslie effect increases between detune zones.
10. Organ I	Triple Drawbar Gospel Organ	Unison - 5th - Suboctave. Leslie effect increases between detune zones.
11. Ensemble	Thin string sound with 'Solina' style 3 phase chorus	Unison - 5th - Suboctave. Lfo Speed increases between detune zones.
12. Noisy Tape	2 Osc Saw modulated by noise in opposite phase and tape effect	Unison - 5th - Suboctave. Noise depth increases between detune zones. Minimum values introduce wow and flutter

	on main output.	onto the main output and the reverb.
13. Lfo X8	Oberheim X8 inspired string sound using 8 independent lfes, one for each voice	Unison - 5th - Suboctave. Lfo Speed increases between detune zones.
14. Lfo Vox	Polymoog Vox Humana inspired. Saw and PWM Square oses modulated by 3 lfes.	Unison - 5th - Suboctave. Lfo Speed increases between detune zones.
15. Supersaw Full	7 Osc Supersaw equally mixed	Unison - 5th - Suboctave.
16. Supersaw Mild	7 Osc Supersaw, Osc 1 is dominant	Unison - 5th - Suboctave.

Effects

Reverb (*reverb time*)

Adjusts the amount of signal sent to the stereo reverb.
Hold shift to adjust reverb time.

Chorus

Hold Shift and Mono/Poly to activate the chorus. The chorus is based on the same effect found on the Ensemble oscillator mode. Turning the detune will increase the speed of the effect. Press Shift and Mono/Poly to toggle chorus on and off.

Inputs and Outputs



Gate In

Triggers each chord in the chord memory, or the last chord played in via midi.

V/Oct

1 volt per octave bipolar CV control of the oscillator pitch.

Filter

CV control of the filter cutoff frequency.

CV A

Assignable CV input A. See [here](#) for details.

CV B

Assignable CV input B. See [here](#) for details.

Midi In

Midi control input for playing the module polyphonically.

Assignable CV

Assigning a CV to a Control Knob

Inserting a patch chord into either CV A or CV B initiates CV assign, the red led will flash rapidly for 2 seconds. Quickly turn any knob within this time and it will be assigned that CV. If you do nothing the CV assignment will not change. You can even assign Shift parameters such as glide, but be quick.

CV levels are added to the current control position so you may need to adjust the knob once assigned.

CV inputs are bipolar so can turn the parameter values up as well as down.

Currently CV A and B can both be assigned to the same parameter with unpredictable results.

Assigning Oscillator Mode

Changing the oscillator mode is possible using a CV. However the mode only changes on new notes, this allows notes already playing to continue however, depending on the incoming and outgoing oscillator modes, lfo amounts may change instantly.

Assigning Filter Frequency

Filter frequency can be assigned to CV A or CV B but remember it has its own dedicated CV input.

Assigning Chord Select

To change chord memory playback via a CV (Chord Select) press the chord **Trigger** button within 2 seconds of inserting a patch cable.

Assigning Pitch Knob

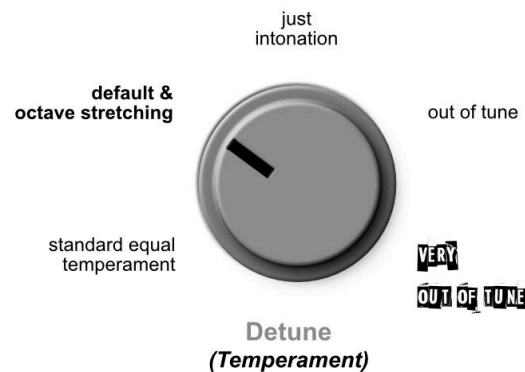
Despite having a dedicated V/Oct input you can assign CV to the Pitch Knob, this will quantise the CV input voltage to the nearest whole note.

Temperament

Monumatic has a very advanced tuning system built into its midi note interpreter.

Unlike most digital (and analogue) keyboards, Monumatic incorporates a self-adjusting temperament called [hermode](#) tuning. This approach deviates from the equal temperament standard which has compromised music over the last century and instead provides the option to adjust the tuning scale to a more natural and more harmonious sound. To hear this in action select a single oscillator sound without any effects and play a three note major chord, set the release to max to make it hold, next hold

Shift and slowly turn the Detune control from left to centre. You should hear the beats of the sound tighten up into a perfectly tuned chord, this is called Just Intonation and with Monumatic you can adjust how much of this effect you want, 12 o'clock being 100%, beyond this will overcompensate and enable out of tune effects. The default setting is at about 10 O'clock on the dial, this default setting also features octave stretching - the effect of sharpening octaves slightly to give a less static 'digital' sounding octave. Used subtly the upshot of this effect is that it gives some added richness to chords that you might find in a string quartet or in a vintage analogue synth. The system itself is constantly seeking the best frequency ratios to trim chords to, micro-tuning notes up or down depending on the arrival of each new note to the chord party. To switch off the system hold Shift and turn Detune fully anti-clockwise. Turning the Temperament control beyond 12 O'clock will overcompensate the intonation correction, sending chords out of tune, useful for more indy sounds. When playing single notes there is no effect.



Updating the Firmware

Even a factory fresh module may not be running the latest Monumatic firmware, so it's always best to update the firmware on your device whenever possible. Any new firmware versions are announced on the Knobula forum and these updates can contain new features, including user feature requests or bugfixes.

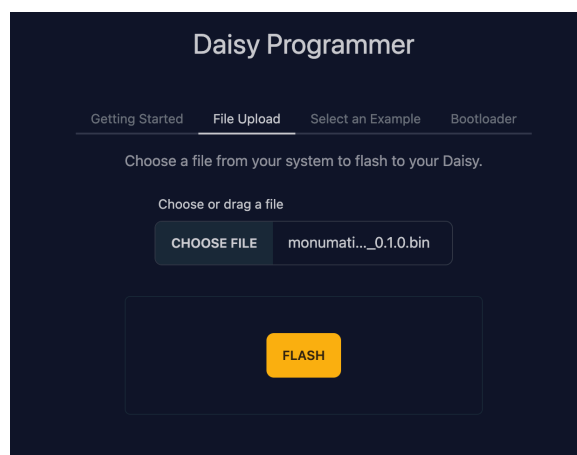
Display Current Firmware Version

To check which firmware version you have installed (introduced in V1.1.4) power up the unit while holding down the mono/poly button. The white trigger LED blinks Major then Minor version (blink - pause - blink for 1.1) then red mono/poly LED blinks Patch version (4 times for .4)

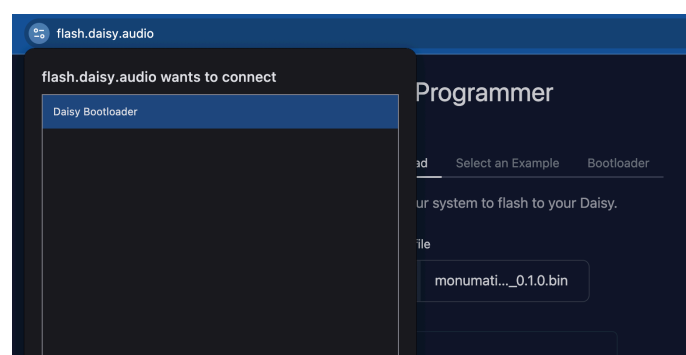
Update Procedure

Updating Firmware on Monumatic - Requires [Chrome](#) browser

1. Download the latest Firmware from the Knobula website [download page](#).
2. Disconnect the ribbon cable and remove Monumatic from your rack.
3. Connect the Micro USB cable from your computer to the Daisy board on the back of the Monumatic.
4. *Press and release the Reset button, then within 2 seconds press and release the Boot button. The led on the back should slowly glow on and off. The unit is now in Boot mode.
5. Open Chrome on your computer and navigate to the [Daisy Programmer](#)
6. Select the File Upload tab.



7. Click the Choose File button, and select the Monumatic.bin file you just downloaded.
8. Click the Flash button, a new window will open asking if you wish to connect to Daisy Boatloader. Click Connect and the Daisy should begin updating.

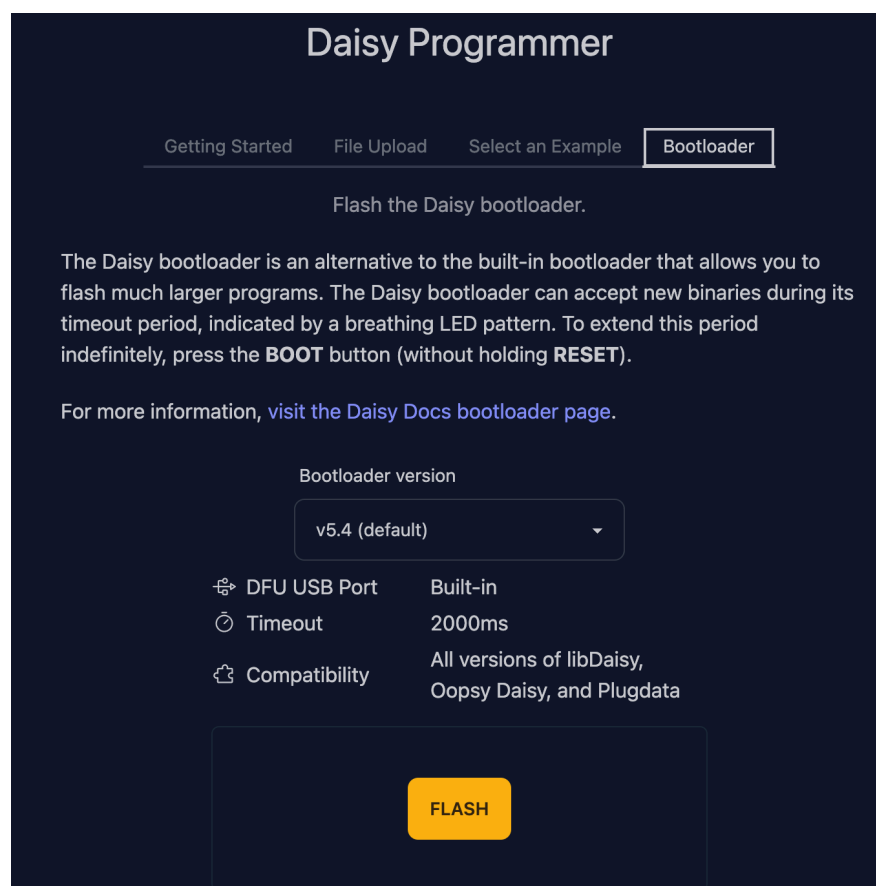


***Ignore the instructions on the Getting Started page. Seasoned Knobula users take note, this module does not update the same way as the Poly Cinematic or Kickain modules, so read carefully.**

Firmware Trouble Shooting

Monumatic No longer working after firmware upgrade

If Monumatic no longer works after a firmware upgrade you may have overwritten the Daisy Boatloader with the firmware by mistake (see step 4 above). If this happens you will need to reinstall the Daisy Bootloader, press the 'Boot' and 'Reset' buttons together, at the same time, simultaneously (do not press separately yet!) and click on the Bootloader tab and flash



from there before returning to the firmware update instructions above.

Monumatic Not showing in Connect Menu

On windows, you may have to update the driver to WinUSB. To do this, you can download the free software, Zadig. Instructions for this can be found [here](#).

CV Calibration

There are 3 ways you can calibrate or reset the CV V/Oct input on Monumatic.

Automatic Calibration

Disconnect any midi or v/oct inputs. Boot up the module while holding down all three buttons; Trigger, Shift and Mono/Poly. Connect a patch cable to the V/Oct input, the Mono / Poly button will begin flashing. Send a 0 volt signal followed by a 1volt signal from a CV keyboard (two notes one octave apart). Monumatic will detect the two voltages and calibrate itself, and the red led will stop. Reboot the module to exit calibration mode.

Note. If a midi message is detected in calibration mode it will emit a loud test tone until all the knobs, switches and sockets have been checked.

Midi CC Calibration

CV range and CV offset can be manually adjusted using Midi CC 23 and Midi CC 24 respectively. Hold Shift and press trigger to save the settings.

Note. Spurious midi CC messages could unintentionally alter these settings if used with certain midi controllers. Midi CC can be disabled with CC 106 and a value greater than 64. Only a factory reset will restore CC control.

Factory Reset

Boot up while holding Shift until the Trigger LED starts flashing. This will reset the whole module to factory settings and chords. It will also set default values for CV Offset and CV Range. Despite the reset this is not a calibrated setting but a useful means to get back to some reasonable values.

Appendix A

Midi CC Chart

Parameter	CC Number	Notes
Glide Rate	5	Sets the time taken to reach a note
Volume	7	
Flip Tuning	21	Values over 64 flip the tuning control. When flipped Fine tuning becomes default the and holding Shift Tune is coarse tuning
Pitch Bend Range	22	Sets the range of the pitch bend
CV Range	23	adjust the volts per octave range
CV Offset	24	adjusts the pitch set at zero volts. V/Oct socket must be connected even for zero volts
Filter CC Amount	42	attenuates the incoming Filter Cutoff
Hold Pedal	64	Value > 99 = Hold, Value < 26 = Release
Glide on/off	65	Values over 64 switch on glide
Osc Mode	70	
Filter Resonance	71	
Release	72	
Attack	73	
Filter Cutoff	74	
Decay	75	
Detune	94	
Filter Env Depth	77	
Pitch Env Depth	80	
Reverb Send	91	
Chorus	93	Values over 64 switch on chorus effect
CC Disable	106	A value above 64 disables CC control. Factory reset to restore control.
Mono/Poly Switch	119	Values over 64 = poly Mode, under 64 = mono mode
Midi Panic	120	Kills all notes playing

WARRANTY

This product comes with a one-year guarantee starting from the product's original purchase date and covers any manufacturing defects or other functional deficiencies that may arise during the warranty period.

The warranty does not apply in case of:

- Damage caused by misuse
- Mechanical damage arising from careless treatment (dropping, vigorous shaking, mishandling, etc.)
- Damage caused by liquids or powders penetrating the device
- Heat damage caused by overexposure to sunlight or heating
- Electric damage caused by improper connection

This warranty covers replacement or repair only where we deem it is reasonable. Please contact us or your dealer via email to obtain an RMA number. Shipping costs of sending a module back for servicing is paid by the customer. Warranty parts and replacements can only be sent to addresses in the country where the item was purchased.

Device complies with all EU regulations concerning RoHS lead-free manufacturing and WEEE disposal.